

Syed Emad Uddin Shubha

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Citizenship: Bangladesh , US Visa Status: F-1 Student

RESEARCH INTEREST

- Quantum Communication
- Quantum Algorithm
- Quantum Hardware Security
- Quantum Error Correction
- Quantum Cryptography
- Deep Learning

EDUCATION

Louisiana State University **January 2025 – Present**
Computer Science Doctoral Student **LSU GPA: 3.988/4.00**
Courses: Quantum Technology (A+), Quantum Algorithms (A+), Machine Learning (A+), Advanced Algorithm Design.
Bangladesh University of Engineering and Technology **February 2017 – May 2022**
B.Sc. in Electrical and Electronic Engineering **CGPA: 3.24/4.00**
Relevant Courses: Control Systems, Solid State Devices, Random Signals & Processes, Communication Systems.

RESEARCH EXPERIENCE

Graduate Research Assistant — Louisiana State University, USA **January 2025 - Present**

- Characterized crosstalk as an active attack vector through initial pulse-level simulations; developed this work into a full pulse-to-circuit framework that identified a novel asymmetric attack mechanism (presented at ACM QSec 2025, co-located with CCS), and extended this methodology to a cross-layer verification and Detection framework for Quantum SoCs. (Currently funded by Louisiana Board of Regents, \$73,926 awarded, Proposal number: 033A-26).
- Designed a correlation-guided adaptive purification protocol for dense coding (presented at QCE 2025). **Drafted research aims, questions, & methodology for an NSF EAGER proposal (\$194,596 awarded, Award number: 2608182).**

Research Assistant — North South University, Dhaka **June 2022 - May 2024**

- Pioneered Bangladesh's first research initiatives in quantum computing and communication. Extended my undergraduate thesis (where I developed a simulation framework to analyze the fidelity of encoded Bell states under noise) into a first-authored high-impact journal article, proposing a novel error correction protocol that uses stabilizer formalism to improve fidelity under Pauli noise channels for both local and long-distance communication.
- Mentored junior researchers; formulated and directed research projects. Research on surface code decoder design was presented at an IEEE conference; integrated QKD with classical security, resulting in two additional journal articles.

JOURNAL PUBLICATIONS

1. "Secure and Efficient n-Qubit Entangled State Teleportation Using Partially Entangled GHZ Channels and Optimal POVM." (**Corresponding author**; accepted in AVS Quantum Science, Q1-indexed, IF: 3.0), **doi: 10.1116/5.0284072**.
2. "Significant improvement of fidelity for encoded quantum bell pairs at long and short-distance communication along with a generalized circuit." (**1st author**; accepted in Heliyon, Q1-indexed, IF: 3.6), **doi: 10.1016/j.heliyon.2023.e19700**.
3. "Multi-Layered Security System: Integrating Quantum Key Distribution with Classical Cryptography to Enhance Steganographic Security". (Co-supervised, accepted in Alexandria Engineering Journal, Q1-indexed, IF=6.8), **doi: 10.1016/j.aej.2025.02.056**.
4. "Enhancing the security of image transmission in Quantum era: a chaos-assisted QKD approach using entanglement." (Co-supervised, accepted in IET Quantum Communication, Q2-indexed, IF=2.8), **doi: 10.1049/qtc2.70016**.

SELECTED PEER-REVIEWED CONFERENCE PROCEEDINGS

5. "Distinguishing Coherent Crosstalk from Calibration Drift via Pauli-Transfer Signatures and Quantum Edge Detection" (**1st author**). Accepted for publication in HOST 2027. **arXiv:2609.40283**.
6. "Pulse-to-Circuit Characterization of Stealthy Crosstalk Attack on Multi-Tenant Superconducting Quantum Hardware" (**1st author**). Presented at ACM Quantum Security Workshop 2025, **doi: 10.1145/3733825.3765282**.
7. "Testbed Protocol for Adaptive Superdense Coding in Distributed Quantum Networks" (**1st author**). Presented at IEEE Quantum Week, 2025. **doi: 10.1109/QCE65121.2025.10350**.

TEACHING EXPERIENCE

Graduate Teaching Assistant — Louisiana State University

August 2025 - Present

- Supporting the instructional activities for CSC 4501 (Computer Networks)
- Grading assignments and providing feedback to students.

Teaching Assistant — BRAC University, Dhaka

June 2023 - December 2024

- Conducted CSE 482 (Quantum Computing II) tutorials and problem-solving classes.
- Conducted lab sessions of MAT120 (Calculus), teaching relevant numerical methods and Python libraries.

NOTABLE PROJECTS

- **Neural Shadows: ML for Entanglement Classification** — [*GitHub Link*](#).
 - Classified entanglement phases in Werner states using an ML pipeline built on sparse classical shadow representations.
 - Quantified classifier robustness (SVM, FNN) against finite-shot noise using Monte Carlo stability analysis.
- **Preparing the W State on Superconducting Qubits Using QISKIT PULSE** — [*GitHub Link*](#).
 - Used real calibration data of ibm_sherbrooke to design necessary gate and measurement pulses to prepare W states.
 - Calibrated π -pulses via Rabi and implemented Echoed Cross-Resonance (CR) to mitigate crosstalk in noisy simulations.
- **Interactive Quantum Protocol Simulations and Curriculum** — [*GitHub Link*](#).
 - Developed interactive modules progressing from basic circuit design to density matrix formalism.
 - Implemented end-to-end simulations and security analysis of B92, BB84, and E91 protocols.
- **Quantum Image Encoding and Decoding using QISKIT** — [*GitHub Link*](#).
 - Implemented quantum image encoding algorithm using FRQI and NEQR method.
 - Generalized both encoding circuit, developed the decoding method and reconstructed the output image.
- **Load Flow Analysis for n Bus using Gauss-Siedel Algorithm** — [*GitHub Link*](#).
 - Developed a load flow analysis program for n-bus power systems during the junior year of undergraduate studies.
 - Implemented generalized Gauss-Siedel algorithm and generated output for user defined tolerance.

SKILLS SUMMARY

Programming Languages : Python, MATLAB, C, C++, Assembly, Rust, QASM.
Quantum Software : Qiskit, QuTiP, Amazon Braket.
Simulation Software : PSpice, Proteus, Quartus, Simulink.
Development & Typesetting: LaTeX, Git.

PROFESSIONAL SERVICE & AFFILIATIONS

- Reviewer, IEEE Transactions on Quantum Engineering (2026).
- Sub-reviewer, IEEE Dallas Circuits and Systems Conference (2026).
- Graduate Student Member, IEEE & ACM (2024–Present).
- Member, The Honor Society of Phi Kappa Phi (Inducted 2026).

HONORS & AWARDS

- Received the **Quantum Rising Star Award** at the Inaugural Quantum Student Success Summit, Rice University. (2026)
- Delivered a lighting talk on crosstalk vulnerabilities in multi-tenant quantum processors at Rice University (2026)
- Recipient of the **NSF Student Travel Grant** (\$1250), to present two first-author papers at QCE 2025.
- Presented at Q-Net Symposium by LSU APS Chapter (2025).
- Instructor at the 3rd PSI-Tensor Winter School for theoretical physics (2024).
- Keynote Speaker for an IEEE NSU Student Branch event on Quantum Computing (2022).
- First Place, University-Level Math Olympiad, organized by Bangladesh Mathematical Society (2021).
- Ranked 47th among 10,000 candidates in the BUET Admission Test, earning University Admission Test Scholarship (2017).

RECOGNITION & MEDIA COVERAGE

- Recognized by the LSU Division of CSE for research contributions at IEEE Quantum Week 2025 — [**Link**](#).
- Acknowledged in research articles published in **AVS Quantum Science**, & **Advanced Quantum Technologies**.
- Research was featured in industry outlet *Quantum Zeitgeist* — [**ACM QSec paper**](#), [**AVS Quantum Science article**](#).

LEADERSHIP & OUTREACH

- Conducted Hands-on Advanced Lab in Qiskit Fall Fest at Louisiana State University.
- Assisted in the distribution of PPE during the COVID-19 emergency.
- Volunteered in the distribution of winter clothes in remote villages.
- Organized a nationwide online course on quantum mechanics and quantum computing.
- Engaged in numerous science outreach initiatives through writing blogs and articles.